Zicong Zhang

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https://zhangzicong518.github.io

Education

Shanghai Jiao Tong University, B.S. in Computer Science Sept 2023 – Present

- John Hopcroft Class, Zhiyuan College (Honor Program)
- GPA: 89.5/100
- **Coursework:**
  Math: Mathematical Logic(97), Information Theory(97), Linear Algebra(94)
  CS: Foundations of Data Science(100), Machine Learning Theory(98), Computer System Design(98), Program Design and Data Structure(96), Machine Learning(95), Algorithm(94)

Research Experience

Jacobian-Aware Post-Training for One-Step Generation, Harvard, intern Aug 2026 – Present

Advisor: Prof. Qianqian Wang

- Started from explorative modeling (best-of-N training); showed that as N scales, the objective becomes theoretically mismatched with the target distribution, causing distribution drift and mode misallocation, which motivated a shift to distribution matching
- Identified generator's Jacobian spectrum as the key bottleneck of amortized transportation. Each update follows the signal with $<1\%$ cosine similarity and post-training rapidly collapses the flat spectrum of generators
- Improved post-training efficiency by reusing training signals (lowering a state-of-the-art method's EMA FID-50k from 1.78 to 1.51) and by incorporating the spectrum regularization to preserve pretrained geometry during post-training

Multimodal Reasoning and Self-Evolving Agent, Microsoft Research Asia, intern July 2025 – Jan 2026

Advisors: Yifan Yang, Chong Luo
Results in paper: EnvOpt: Closed-Loop Environment Optimization for Agent Post-Training

- Designed a tool-integrated RL paradigm that unifies long-chain-of-thought reasoning and tool-calling into a hybrid thinking mode, enabling vision–language models to "think with images" more effectively
- Built **EnvOpt**, a closed-loop framework that diagnoses target-model behavior, connects training targets to realized RL outcomes and subsequent evaluation, and iteratively constructs executable environments faithful to the intended target

Continual Learning with Model Fusion, SJTU MIFA Lab, intern July 2024 – April 2025

Advisor: Prof. Weiran Huang
Results in paper: Enhanced Continual Learning of Vision-Language Models with Model Fusion

- Co-led development of a novel continual learning framework for VLMs by introducing model fusion
- Proposed aggregating the results of multiple decoupled task-specific models for prediction in zero-shot scenarios
- Deployed the full experiment pipeline and conducted extensive experiments on multiple benchmarks, consistently outperforming the original pre-trained VLM and other state-of-the-art continual learning methods by up to 2%

Awards

- Zhiyuan Honor Awards (Top 10% in SJTU) 2023,2024, 2025. 2026
- The Third Prize of Academic Scholarship (Top 30% in major) 2024, 2025, 2026
- Honorable Mention, Mathematical Contest In Modeling 2025

Publications

Enhanced Continual Learning of Vision-Language Models with Model Fusion

Haoyuan Gao*, Zicong Zhang*, Yuqi Wei, Linglan Zhao, Guilin Li, Yexin Li, Linghe Kong, Weiran Huang

The Fourteenth International Conference on Learning Representations (ICLR2026)

XSearch: Explainable Code Search via Concept-to-Code Alignment

Yiming Liu, Ruofan Liu, Yun Lin, Zicong Zhang, Weiyu Kong, Pengnian Qi, Xiao Cheng, Weinan Zhang, Qianxiang Wang, Linpeng Huang

The 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA2026)

EnvOpt: Closed-Loop Environment Optimization for Agent Post-Training

Zicong Zhang*, Zisu Huang*, Jingwen Xu, Rui Wang, Yifan Yang, Muzhao Tian, Xuemei Gao, Qi Dai, Bei Liu, Kai Qiu, Dongdong Chen, Xiaoqing Zheng, Chong Luo

[Under Review]

Projects

Environment Optimization for Self-Evolving Agent – *Microsoft Research Asia* Dec 2025 – July 2026

Results in paper: EnvOpt: Closed-Loop Environment Optimization for Agent Post-Training

- Diagnosed three failure modes of agent-as-data-researcher loops: incomplete behavioral analysis, disconnected cross-round feedback (only 21% of generated tasks provide learnable reward signal), and unfaithful realization of executable environments
- Designed the corresponding mechanisms—traceable behavioral map, target-conditioned context refinement, and iterative environment construction—raising the learnable-task rate from 21% to 57% and producing environments more reliable and better aligned with training targets
- Achieved $\sim$+15 pp average improvement over the base model across four agentic tool-use benchmarks, substantially outperforming both a vanilla autonomous coding-agent researcher and a fixed-environment baseline

Scaling Active Perception for Multimodal Reasoning – *Microsoft Research Asia* July 2025 – Jan 2026

- Unified long-chain reasoning and tool-calling into a single hybrid thinking mode via SFT and RL, enabling vision-language models to autonomously switch thinking modes while "thinking with images"
- Designed mode-aware process rewards that encourage the model to leverage internal reasoning and external tools in a complementary manner
- Surpassed the original VLM baseline on math and perception-oriented benchmarks including MathVision, MathVista, HRBench, and V*

Technologies

Language: Mandarin (native), English (CET-6 600, TOEFL under preparation)

Programming: C/C++, Python, Rust

Technologies: Git, Pytorch, Latex, vLLM, VeRL, LLaMA-Factory, Deepspeed